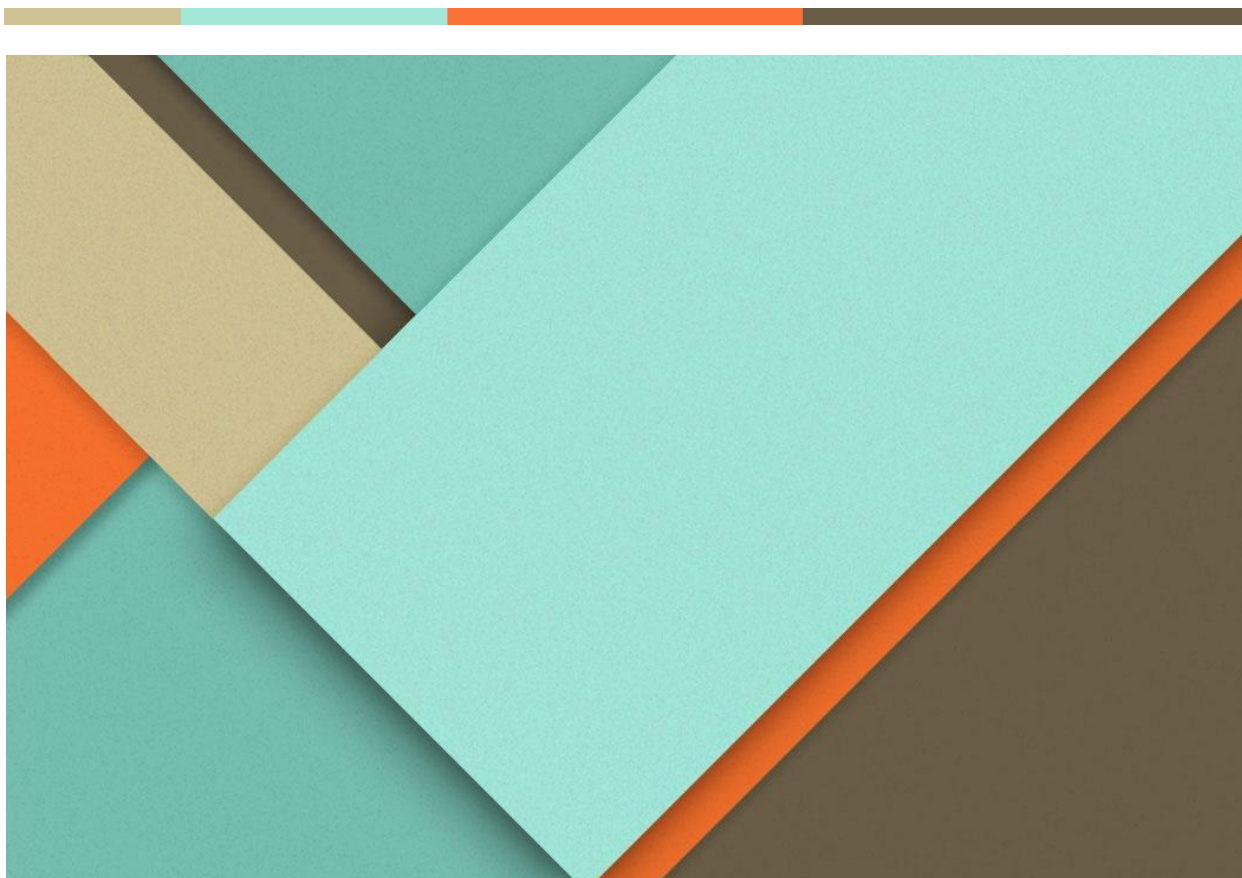




پروژه دوم شبکه

کسرا نوربخش ۸۱۰۱۰۰۲۳۰

سید مهدی سید حاجی حسین ۸۱۰۱۰۰۱۱۸

مقدمه

در این پروژه برخی از مفاهیم CCNA بررسی شدند که از نرم افزار 3GNS استفاده کردیم و توپولوژی های مختلف را پیاده و تست نمودیم. برای سوییچ از خود سوییچ داخل نرم افزار استفاده نمودیم و برای روتر هم از لینک داده شده یک IOS Image دانلود کردیم که در فایل گذاشتیم (1c7200-adventerprisek9-mz.122-33.SRC). در این پروژه هر کدام از بخش ها را در یک دیوایس و به کمک همدیگر انجام دادیم.

گام اول

در این بخش IP های مختلف به ۴ کامپیوتر با استفاده از دستور ip دادیم و سپس از هر کدام با استفاده از ping گرفتن ارتباطشان را بررسی نمودیم که در عکس های پایین نتایج و IP های داده شده مشخص است:



```
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC2       192.168.1.2/24  0.0.0.0      00:50:79:66:68:01  10014  127.0.0.1:10015
          fe80::250:79ff:fe66:6801/64

PC2>
```

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عکس ۳ : IP کامپیوتر دوم

```
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC3       192.168.1.3/24  0.0.0.0      00:50:79:66:68:02  10016  127.0.0.1:10017
          fe80::250:79ff:fe66:6802/64

PC3>
```

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عکس ۴ : IP کامپیوتر سوم

```

PC4
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC4       192.168.1.4/24  0.0.0.0      00:50:79:66:68:03  10018  127.0.0.1:10019
          fe80::250:79ff:fe66:6803/64
PC4>

```

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عکس ۵: IP کامپیوتر چهارم

```

PC1
ping 192.168.1.2/24
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=2.067 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=2.645 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.962 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=2.034 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=2.587 ms

PC1> ping 192.168.1.4/24
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=3.470 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=5.905 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.400 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=5.385 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=3.122 ms

PC1> ping 192.168.1.10/24
host (192.168.1.10) not reachable

PC1>

```

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عکس ۶: تست ping از کامپیوتر اول به دومی و چهارمی و یک IP ست نشده

```

PC4
ping 192.168.1.3/24
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.724 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=3.105 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=3.578 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.132 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.210 ms

PC4> ping 192.168.1.2/24
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=2.337 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=4.684 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=3.095 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=2.367 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=2.365 ms

PC4> ping 192.168.1.5/24
host (192.168.1.5) not reachable

PC4>

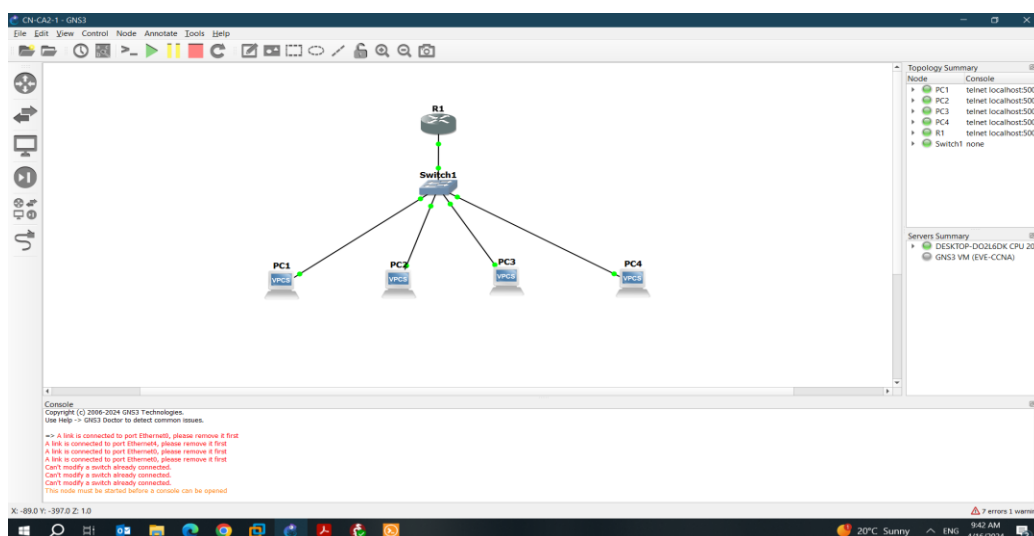
```

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عکس ۷ : تست ping از کامپیوتر چهارم به دومی و سومی و یک IP ست نشده

گام دوم

در این بخش ابتدا IP ها را در رنج داده شده دادیم و سپس Vlan ها را در بخش configure سوییچ تعریف کردیم و آن پورت هایی از سوییچ را که متصل به Vlan مربوطه هستند ست کردیم. برای آن که ارتباط میان Vlan 2 برقرار شود هم sub interface 2 تعریف کردیم و برای هر کدام ip دادیم که این ip را به عنوان gateway کامپیوتر ها گذاشتیم. نتایج در عکس های پایین آمده است:



عکس ۸ : توپولوژی بخش ۲

```
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC1       192.168.10.1/24  0.0.0.0     00:50:79:66:68:00  20012  127.0.0.1:20013
fe80::250:79ff:fe66:6800/64

PC1>
```

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عکس ۹ : IP کامپیوتر اول

```
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC2       192.168.10.2/24  0.0.0.0     00:50:79:66:68:01  20014  127.0.0.1:20015
fe80::250:79ff:fe66:6801/64

PC2>
```

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عکس ۱۰ : IP کامپیوتر دوم

```

PC3
show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC3       192.168.20.1/24  0.0.0.0      00:50:79:66:68:02  20016  127.0.0.1:20017
          fe80::250:79ff:fe66:6802/64

PC3>

```

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عکس ۱۱ : IP کامپیوتر سوم

```

PC4
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC4> ip 192.168.20.2/24
Checking for duplicate address...
PC1 : 192.168.20.2 255.255.255.0

PC4> show
NAME      IP/MASK      GATEWAY      MAC          LPORT  RHOST:PORT
PC4       192.168.20.2/24  0.0.0.0      00:50:79:66:68:03  20018  127.0.0.1:20019
          fe80::250:79ff:fe66:6803/64

PC4>

```

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عکس ۱۲ : IP کامپیوتر چهارم

```

PC1
ping 192.168.10.2/24
84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=3.874 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=2.252 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=1.696 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=3.255 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=64 time=3.163 ms

PC1> ping 192.168.20.1/24
No gateway found

PC1> ping 192.168.20.2/24
No gateway found

PC1> ping 192.168.10.5/24
host (192.168.10.5) not reachable

PC1>

```

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عکس ۱۳ : تست ping از کامپیوتر اول به دومی و سومی و چهارمی و یک IP ست نشده

```

PC2
ping 192.168.10.1/24
84 bytes from 192.168.10.1 icmp_seq=1 ttl=64 time=1.538 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=64 time=2.019 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=64 time=1.251 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=64 time=1.389 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=64 time=1.931 ms

PC2> ping 192.168.20.1/24
No gateway found

PC2> ping 192.168.20.2/24
No gateway found

PC2> ping 192.168.10.4/24
host (192.168.10.4) not reachable

PC2>

```

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عکس ۱۴ : تست ping از کامپیوتر دومی به اولی و سومی و چهارمی و یک IP ست نشده


```

ping 192.168.20.1/24
84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=1.855 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=1.915 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=2.998 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=0.839 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=1.898 ms

PC4> ping 192.168.10.1/24
No gateway found

PC4> ping 192.168.10.2/24
No gateway found

PC4> ping 192.168.20.5/24
host (192.168.20.5) not reachable

PC4>

```

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عکس ۱۵ : تست ping از کامپیوتر چهارمی به سومی و اولی و دومی و یک IP ست نشده

```

ping 192.168.20.2/24
84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=1.245 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=2.478 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=1.120 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=1.412 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=1.834 ms

PC3> ping 192.168.10.2/24
No gateway found

PC3> ping 192.168.10.1/24
No gateway found

PC3> ping 192.168.20.3/24
host (192.168.20.3) not reachable

PC3>

```

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عکس ۱۶ : تست ping از کامپیوتر سومی به چهارمی و دومی و اولی و یک IP ست نشده

```

R1
R1(config)#interface fastEthernet0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#
R1#ip ro
*Apr 16 09:35:16.319: %SYS-5-CONFIG_I: Configured from console by console
R1#ip routing
^
% Invalid input detected at '^' marker.

R1#conf
R1#configure ter
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip routing
R1(config)#end
R1#
*Apr 16 09:35:33.859: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0.10
L       192.168.10.1/32 is directly connected, FastEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, FastEthernet0/0.20
L       192.168.20.1/32 is directly connected, FastEthernet0/0.20
R1#

```

عکس ۱۷ : جدول route روتر

Switch1_Ethernet1_to_PC1_Ethernet0.pcap									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
Apply a display filter ... <Ctrl-/>									
No.	Time	Source	Destination	Protocol	Length	Info			
1	0.000000	00:50:79:66:68:00	Broadcast	ARP	64	Who has 192.168.10.3? Tell 192.168.10.1			
2	0.998943	00:50:79:66:68:00	Broadcast	ARP	64	Who has 192.168.10.3? Tell 192.168.10.1			
3	2.012769	00:50:79:66:68:00	Broadcast	ARP	64	Who has 192.168.10.3? Tell 192.168.10.1			
4	7.102212	00:50:79:66:68:00	Broadcast	ARP	64	Who has 192.168.10.2? Tell 192.168.10.1			
5	7.103204	00:50:79:66:68:01	00:50:79:66:68:00	ARP	64	192.168.10.2 is at 00:50:79:66:68:01			
6	7.117589	192.168.10.1	192.168.10.2	ICMP	98	Echo (ping) request id=0x6ee4, seq=1/256, ttl=64 (reply in 7)			
7	7.118585	192.168.10.2	192.168.10.1	ICMP	98	Echo (ping) reply id=0x6ee4, seq=1/256, ttl=64 (request in 6)			
8	8.147783	192.168.10.1	192.168.10.2	ICMP	98	Echo (ping) request id=0x6fe4, seq=2/512, ttl=64 (reply in 9)			
9	8.148773	192.168.10.2	192.168.10.1	ICMP	98	Echo (ping) reply id=0x6fe4, seq=2/512, ttl=64 (request in 8)			
10	9.177969	192.168.10.1	192.168.10.2	ICMP	98	Echo (ping) request id=0x70e4, seq=3/768, ttl=64 (reply in 11)			
11	9.178999	192.168.10.2	192.168.10.1	ICMP	98	Echo (ping) reply id=0x70e4, seq=3/768, ttl=64 (request in 10)			
12	10.208164	192.168.10.1	192.168.10.2	ICMP	98	Echo (ping) request id=0x71e4, seq=4/1024, ttl=64 (reply in 13)			
13	10.209155	192.168.10.2	192.168.10.1	ICMP	98	Echo (ping) reply id=0x71e4, seq=4/1024, ttl=64 (request in 12)			
14	11.238351	192.168.10.1	192.168.10.2	ICMP	98	Echo (ping) request id=0x72e4, seq=5/1280, ttl=64 (reply in 15)			
15	11.239346	192.168.10.2	192.168.10.1	ICMP	98	Echo (ping) reply id=0x72e4, seq=5/1280, ttl=64 (request in 14)			

> Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits)	0000	ff ff ff ff ff ff ff ff	79 66 68 00 08 06 00 01	P yfh.....
> Ethernet II, Src: 00:50:79:66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010	08 00 06 04 00 01 00 50	79 66 68 00 c0 a8 0a 01	P yfh.....
> Address Resolution Protocol (request)	0020	ff ff ff ff ff ff c0 a8	0a 03 00 00 00 00 00 00	
	0030	aa aa aa aa aa aa aa aa	aa aa aa aa aa aa aa aa	

عکس ۱۸ : همان طور که از تصویر پیداست ارتباط را در 10 VLAN با استفاده از wireshark تست نمودیم که پکت های ICMP نشان داده شده حاصل پینگ های گرفته شده است

چون ابتدا آی پی های داده شده به کامپیوتر ها gateway نداشت پینگ ها به vlan های دیگر نمیرفت که با دادن gateway درست شد. نتیجه پینگ های نهایی :

```

PC1
ping 192.168.10.1
192.168.10.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.10.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.10.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.10.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.10.1 icmp_seq=5 ttl=64 time=0.001 ms

PC3> ping 192.168.10.2
84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=1.860 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=1.916 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=1.887 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=1.936 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=64 time=1.551 ms

PC3> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=77.330 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=62.070 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=62.109 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=61.888 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=61.852 ms

PC3> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=63 time=92.554 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=63 time=61.998 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=63 time=62.114 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=63 time=61.962 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=63 time=62.025 ms

PC3>

```

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گام سوم

در این بخش ابتدا برای آن که بتوانیم از طریق پورت سریال روتر ها را بهم وصل کنیم به هر کدام یک مایکروسافت PA-4T+ اضافه کردیم. ابتدا در رنج های گفته شده IP دادیم و سپس با استفاده از دستور ip route مسیر ها را برای تک تک روتر ها مشخص کردیم و در نهایت هم ارتباط مابین روتر سمت چپ و پایین را قطع کردیم و دوباره تست گرفتیم. نتایج این بخش:

```
PC1
ping 192.168.1.1
192.168.1.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=5 ttl=64 time=0.001 ms

PC1> ping 192.168.4.1
84 bytes from 192.168.4.1 icmp_seq=1 ttl=61 time=185.055 ms
84 bytes from 192.168.4.1 icmp_seq=2 ttl=61 time=93.149 ms
84 bytes from 192.168.4.1 icmp_seq=3 ttl=61 time=108.747 ms
84 bytes from 192.168.4.1 icmp_seq=4 ttl=61 time=108.312 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=61 time=123.587 ms

PC1>
```

عکس ۱۹ : ارتباط بین ۲ کامپیوتر

```
PC2
ping 192.168.4.1
192.168.4.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=5 ttl=64 time=0.001 ms

PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=61 time=108.219 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=61 time=107.979 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=61 time=92.791 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=61 time=108.003 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=61 time=108.716 ms

PC2>
```

عکس ۲۰ : ارتباط بین ۲ کامپیوتر

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial1/0
L    10.0.0.1/32 is directly connected, Serial1/0
S    10.0.0.4/30 [1/0] via 10.0.0.2
C    10.0.0.8/30 is directly connected, Serial1/1
L    10.0.0.9/32 is directly connected, Serial1/1
S    10.0.0.12/30 [1/0] via 10.0.0.2
C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.1.2/32 is directly connected, FastEthernet0/0
S    192.168.4.0/24 [1/0] via 10.0.0.2
R1#

```

عکس ۲۱: جدول مسیر های روتر اول

```

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial1/0
L    10.0.0.2/32 is directly connected, Serial1/0
C    10.0.0.4/30 is directly connected, Serial1/1
L    10.0.0.5/32 is directly connected, Serial1/1
S    10.0.0.8/30 [1/0] via 10.0.0.6
S    10.0.0.12/30 [1/0] via 10.0.0.6
S    192.168.1.0/24 [1/0] via 10.0.0.1
S    192.168.4.0/24 [1/0] via 10.0.0.6
R2#

```

عکس ۲۲: جدول مسیر های روتر دوم

```

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
S    10.0.0.0/30 [1/0] via 10.0.0.14
S    10.0.0.4/30 [1/0] via 10.0.0.14
C    10.0.0.8/30 is directly connected, Serial1/0
L    10.0.0.10/32 is directly connected, Serial1/0
C    10.0.0.12/30 is directly connected, Serial1/1
L    10.0.0.13/32 is directly connected, Serial1/1
S    192.168.1.0/24 [1/0] via 10.0.0.14
S    192.168.4.0/24 [1/0] via 10.0.0.14
R3#

```

عکس ۲۳: جدول مسیر های روتر سوم

```

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
S    10.0.0.0/30 [1/0] via 10.0.0.5
C    10.0.0.4/30 is directly connected, Serial1/0
L    10.0.0.6/32 is directly connected, Serial1/0
S    10.0.0.8/30 [1/0] via 10.0.0.5
C    10.0.0.12/30 is directly connected, Serial1/1
L    10.0.0.14/32 is directly connected, Serial1/1
S    192.168.1.0/24 [1/0] via 10.0.0.5
C    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, FastEthernet0/0
L    192.168.4.2/32 is directly connected, FastEthernet0/0
R4#

```

عکس ۲۴: جدول مسیر های روتر چهارم

```

PC1
ping 192.168.1.1
192.168.1.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.1 icmp_seq=5 ttl=64 time=0.001 ms

PC1> ping 192.168.4.1
192.168.4.1 icmp_seq=1 timeout
192.168.4.1 icmp_seq=2 timeout
84 bytes from 192.168.4.1 icmp_seq=3 ttl=61 time=107.795 ms
84 bytes from 192.168.4.1 icmp_seq=4 ttl=61 time=123.311 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=61 time=92.826 ms

PC1>

```

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عکس ۲۵ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر

```

PC2
ping 192.168.4.1
192.168.4.1 icmp_seq=1 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=2 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=3 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=4 ttl=64 time=0.001 ms
192.168.4.1 icmp_seq=5 ttl=64 time=0.001 ms

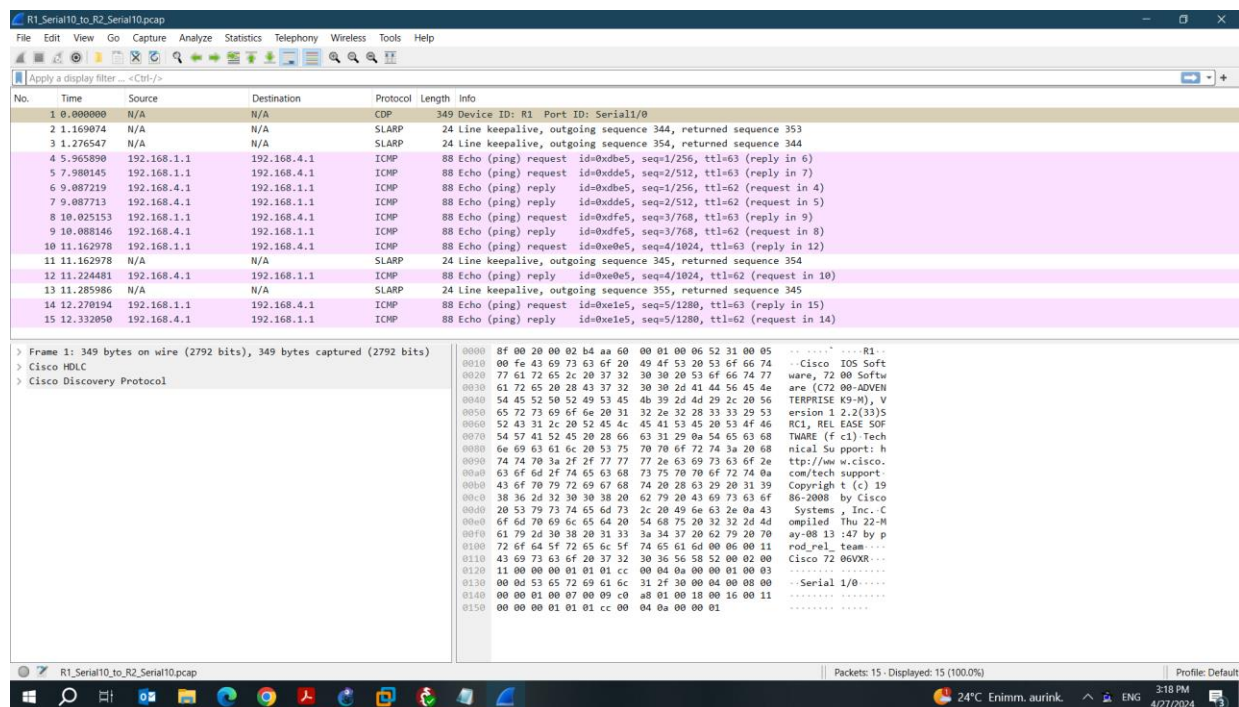
PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=61 time=107.948 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=61 time=92.869 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=61 time=107.317 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=61 time=92.769 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=61 time=123.248 ms

PC2>

```

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عکس ۲۶ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر



عکس ۲۷ : ارتباط بین ۲ کامپیوتر بعد از قطع کردن لینک بین ۲ روتر در وایرشارک

گام چهارم

در این بخش ابتدا برای آن که بتوانیم از طریق پورت سریال روتر ها را بهم وصل کنیم به هر کدام یک ماژول PA-4T+ اضافه کردیم. ابتدا در رنج های گفته شده IP دادیم و سپس با استفاده از دستور router ospf و network مسیر ها را برای تک تک روتر ها مشخص کردیم و در نهایت هم تست گرفتیم. نتایج این بخش:

```

R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
O       192.168.3.0/24 [110/129] via 192.168.14.2, 00:04:30, Serial1/1
        [110/129] via 192.168.12.2, 00:04:30, Serial1/0
    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, Serial1/0
L       192.168.12.1/32 is directly connected, Serial1/0
    192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.14.0/24 is directly connected, Serial1/1
L       192.168.14.1/32 is directly connected, Serial1/1
O       192.168.23.0/24 [110/128] via 192.168.12.2, 00:07:03, Serial1/0
O       192.168.34.0/24 [110/128] via 192.168.14.2, 00:07:03, Serial1/1
R3#show ip ospf nei

Neighbor ID    Pri  State           Dead Time   Address        Interface
192.168.34.1    0    FULL/ -         00:00:36    192.168.14.2   Serial1/1
192.168.23.1    0    FULL/ -         00:00:32    192.168.12.2   Serial1/0
R3#show ip ospf
Routing Process "ospf 3" with ID 192.168.14.1
Start time: 01:30:41.436, Time elapsed: 00:17:11.052
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)

```

عکس 28 : اطلاعات روتر 3

```

R3#
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPF's 10000 msecs
Maximum wait time between two consecutive SPF's 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0, Checksum Sum 0x000000
Number of opaque AS LSA 0, Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1, 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
    Number of interfaces in this area is 3
    Area has no authentication
    SPF algorithm last executed 00:05:05.980 ago
    SPF algorithm executed 7 times
    Area ranges are
    Number of LSA 4, Checksum Sum 0x01C211
    Number of opaque link LSA 0, Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

عکس 29 : ادامه اطلاعات روتر 3


```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O    192.168.1.0/24 [110/65] via 192.168.12.1, 00:17:47, Serial1/0
O    192.168.3.0/24 [110/65] via 192.168.23.2, 00:05:54, Serial1/1
O    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, Serial1/0
L    192.168.12.2/32 is directly connected, Serial1/0
O    192.168.14.0/24 [110/128] via 192.168.12.1, 00:17:57, Serial1/0
O    192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, Serial1/1
L    192.168.23.1/32 is directly connected, Serial1/1
O    192.168.34.0/24 [110/128] via 192.168.23.2, 00:17:37, Serial1/1
R1#show ospf nei
% Invalid input detected at '^' marker.

R1#show ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.34.2      0    FULL/ -         00:00:38    192.168.23.2   Serial1/1
192.168.14.1      0    FULL/ -         00:00:34    192.168.12.1   Serial1/0
R1#show ip ospf
Routing Process "ospf 1" with ID 192.168.23.1
Start time: 01:29:24.456, Time elapsed: 00:19:26.248
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)

```

عکس 30 : اطلاعات روتر 1

```

R1#
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPF's 10000 msecs
Maximum wait time between two consecutive SPF's 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
  Number of interfaces in this area is 2
  Area has no authentication
  SPF algorithm last executed 00:06:12.348 ago
  SPF algorithm executed 8 times
  Area ranges are
    Number of LSA 4. Checksum Sum 0x01C211
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
R1#

```

عکس 31 : ادامه اطلاعات روتر 1

```

R2#
R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O    192.168.1.0/24 [110/65] via 192.168.14.1, 00:18:28, Serial1/0
O    192.168.3.0/24 [110/65] via 192.168.34.2, 00:06:31, Serial1/1
O    192.168.12.0/24 [110/128] via 192.168.14.1, 00:18:28, Serial1/0
O    192.168.14.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.14.0/24 is directly connected, Serial1/0
L     192.168.14.2/32 is directly connected, Serial1/0
O    192.168.23.0/24 [110/128] via 192.168.34.2, 00:18:13, Serial1/1
O    192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.34.0/24 is directly connected, Serial1/1
L     192.168.34.1/32 is directly connected, Serial1/1
R2#show ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.34.2      0    FULL/ -         00:00:36    192.168.34.2   Serial1/1
192.168.14.1      0    FULL/ -         00:00:33    192.168.14.1   Serial1/0
R2#show ip ospf
Routing Process "ospf 2" with ID 192.168.34.1
Start time: 01:30:15.776, Time elapsed: 00:19:11.848
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic

```

عکس 32 : اطلاعات روتر 2

```

R2#
R2#
R2#show ip ospf nei
Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.34.2      0    FULL/ -         00:00:36    192.168.34.2   Serial1/1
192.168.14.1      0    FULL/ -         00:00:33    192.168.14.1   Serial1/0
R2#show ip ospf
Routing Process "ospf 2" with ID 192.168.34.1
Start time: 01:30:15.776, Time elapsed: 00:19:11.848
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPF's 10000 msecs
Maximum wait time between two consecutive SPF's 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
R2#

```

عکس 33 : ادامه اطلاعات روتر 2

```

R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

O    192.168.1.0/24 [110/129] via 192.168.34.1, 00:09:03, Serial1/1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, FastEthernet0/0
L    192.168.3.1/32 is directly connected, FastEthernet0/0
O    192.168.12.0/24 [110/128] via 192.168.23.1, 00:09:03, Serial1/0
O    192.168.14.0/24 [110/128] via 192.168.34.1, 00:09:03, Serial1/1
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.23.0/24 is directly connected, Serial1/0
L    192.168.23.2/32 is directly connected, Serial1/0
     192.168.34.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.34.0/24 is directly connected, Serial1/1
L    192.168.34.2/32 is directly connected, Serial1/1
R4#show ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.34.1      0    FULL/ -         00:00:32    192.168.34.1   Serial1/1
192.168.23.1      0    FULL/ -         00:00:31    192.168.23.1   Serial1/0
R4#show ip ospf
Routing Process "ospf 4" with ID 192.168.34.2
Start time: 01:31:01.648, Time elapsed: 00:19:09.456
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)

```

عکس 34 : اطلاعات روتر 4

```

Supports area transit capability
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 5000 msecs
Minimum hold time between two consecutive SPF's 10000 msecs
Maximum wait time between two consecutive SPF's 10000 msecs
Incremental-SPF disabled
Minimum LSA interval 5 secs
Minimum LSA arrival 1000 msecs
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msecs
Retransmission pacing timer 66 msecs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
  Number of interfaces in this area is 3
  Area has no authentication
  SPF algorithm last executed 00:07:23.644 ago
  SPF algorithm executed 4 times
  Area ranges are
    Number of LSA 4. Checksum Sum 0x01C211
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0
R4#

```

عکس 35 : ادامه اطلاعات روتر 4

```
PC1
ping 192.168.1.2
192.168.1.2 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.2 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.2 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.2 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.2 icmp_seq=5 ttl=64 time=0.001 ms

PC1> ping 192.168.3.2
192.168.3.2 icmp_seq=1 timeout
192.168.3.2 icmp_seq=2 timeout
84 bytes from 192.168.3.2 icmp_seq=3 ttl=61 time=90.290 ms
84 bytes from 192.168.3.2 icmp_seq=4 ttl=61 time=106.087 ms
84 bytes from 192.168.3.2 icmp_seq=5 ttl=61 time=105.276 ms

PC1> 
```

عکس 36 : ارتباط بین کامپیوتر اولی به دومی

```
PC2
ping 192.168.3.2
192.168.3.2 icmp_seq=1 ttl=64 time=0.001 ms
192.168.3.2 icmp_seq=2 ttl=64 time=0.001 ms
192.168.3.2 icmp_seq=3 ttl=64 time=0.001 ms
192.168.3.2 icmp_seq=4 ttl=64 time=0.001 ms
192.168.3.2 icmp_seq=5 ttl=64 time=0.001 ms

PC2> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=61 time=105.002 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=61 time=90.346 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=61 time=120.628 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=61 time=104.921 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=61 time=119.919 ms

PC2> 
```

عکس 37 : ارتباط بین کامپیوتر دومی به اولی

The image shows a Wireshark packet capture window titled "Switch1_Ethernet0_to_R3_FastEthernet00.pcap". The interface includes a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help), a toolbar, and a display filter set to "<Ctrl-/>".

The packet list pane shows 16 captured packets. The first 16 packets are summarized in the table below:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ca:03:0f:b4:00:00	ca:03:0f:b4:00:00	LOOP	60	Reply
2	4.208426	192.168.1.1	224.0.0.5	OSPF	90	Hello Packet
3	6.769624	192.168.1.2	192.168.3.2	ICMP	98	Echo (ping) request id=0x2909, seq=1/256, ttl=64 (reply in 4)
4	6.860007	192.168.3.2	192.168.1.2	ICMP	98	Echo (ping) reply id=0x2909, seq=1/256, ttl=61 (request in 3)
5	7.877707	192.168.1.2	192.168.3.2	ICMP	98	Echo (ping) request id=0x2b09, seq=2/512, ttl=64 (reply in 6)
6	7.997390	192.168.3.2	192.168.1.2	ICMP	98	Echo (ping) reply id=0x2b09, seq=2/512, ttl=61 (request in 5)
7	9.017085	192.168.1.2	192.168.3.2	ICMP	98	Echo (ping) request id=0x2c09, seq=3/768, ttl=64 (reply in 8)
8	9.135766	192.168.3.2	192.168.1.2	ICMP	98	Echo (ping) reply id=0x2c09, seq=3/768, ttl=61 (request in 7)
9	10.003992	ca:03:0f:b4:00:00	ca:03:0f:b4:00:00	LOOP	60	Reply
10	10.153590	192.168.1.2	192.168.3.2	ICMP	98	Echo (ping) request id=0x2d09, seq=4/1024, ttl=64 (reply in 11)
11	10.243351	192.168.3.2	192.168.1.2	ICMP	98	Echo (ping) reply id=0x2d09, seq=4/1024, ttl=61 (request in 10)
12	11.261884	192.168.1.2	192.168.3.2	ICMP	98	Echo (ping) request id=0x2e09, seq=5/1280, ttl=64 (reply in 13)
13	11.366601	192.168.3.2	192.168.1.2	ICMP	98	Echo (ping) reply id=0x2e09, seq=5/1280, ttl=61 (request in 12)
14	13.333011	192.168.1.1	224.0.0.5	OSPF	90	Hello Packet
15	19.987367	ca:03:0f:b4:00:00	ca:03:0f:b4:00:00	LOOP	60	Reply
16	23.067475	192.168.1.1	224.0.0.5	OSPF	90	Hello Packet

The packet details pane for packet 16 shows the following structure:

- Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
- Ethernet II, Src: ca:03:0f:b4:00:00 (ca:03:0f:b4:00:00), Dst: ca:03:0f:b4:00:00
- Configuration Test Protocol (loopback)
- Data (40 bytes)

The packet bytes pane shows the raw data in hexadecimal and ASCII format.

The status bar at the bottom indicates "Packets: 18 · Displayed: 18 (100.0%)" and "Profile: Default". The system tray shows the date and time as 4/27/2024, 5:48 PM, and the weather as 25°C Sunny.

عکس 38: ارتباط بین دو کامپیوتر در وایرشارک